Solutions to JEE Main - 8 | JEE-2022

PHYSICS

SECTION-1

1.(B) Conserving energy between the initial and final instants,

$$K + mgH = 2K \quad \Rightarrow \quad mgH = K$$

Let the kinetic energy of the particle when it is at a height h above the ground be denoted by $K(h)$

Now, conserving energy between the initial instant and the instant when the particle is at a height h above the ground,

$$K + mgH = K(h) + mgh \;\Rightarrow\; K(h) = K + mgH - mgh = K + K - K\left(\frac{h}{H}\right) = K\left(2 - \frac{h}{H}\right)$$

2.(C) Let the distance of the axis P from the centre be x

Then, by parallel axis theorem, $I_P = I_{CM} + mx^2 \quad \Rightarrow \quad I_P = \frac{2}{5}mR^2 + mx^2$

We are given that $I_P = mR^2$

Therefore, $\frac{2}{5}mR^2 + mx^2 = mR^2 \quad \Rightarrow \quad x = \sqrt{\frac{3}{5}}R \approx 0.78R$

3.(C) Conserving energy between the surface and height R above the surface,

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = \frac{1}{2}m\left(\frac{v}{4}\right)^2 - \frac{GMm}{2R} \quad \Rightarrow \quad v = \sqrt{\frac{16GM}{15R}}$$

Let the maximum height above the surface that the object reaches be h

Then, conserving energy between the surface and the maximum height,

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = -\frac{GMm}{R+h} \quad \Rightarrow \quad \frac{8GM}{15R} - \frac{GM}{R} = -\frac{GM}{R+h} \quad \Rightarrow \quad h = \frac{8R}{7}$$

4.(A) Fundamental frequency of the segment AB, $f_0 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$

We are given that $T = 75\text{ N}$ $\quad \mu = 0.03\text{ kg/m}$ $\quad f_0 = 300\text{ Hz}$

Solving, we get $L = \frac{1}{6}\text{ m} = 16.7\text{ cm}$

5.(C) Weight of the sphere in air, $W_A = \left(\frac{4\pi}{3}(2)^3(3) + \frac{4\pi}{3}(5^3 - 2^3)(0.5)\right)(10) = \frac{4\pi}{3}(825)$

Let the fraction of the volume of the sphere that is immersed in water be x

Buoyant force acting on the sphere, $B = \left(x\left(\frac{4\pi}{3}(5)^3\right)\right)(1)(10) = \frac{4\pi}{3}(1250)x$

In equilibrium, $W_A = B \quad \Rightarrow \quad \frac{4\pi}{3}(825) = \frac{4\pi}{3}(1250)x \quad \Rightarrow \quad x = 0.66$

6.(A) We know that terminal velocity is $v_T = \frac{2(\rho - \rho_L)r^2 g}{9\eta}$

7.(B) Let the magnitudes of the forces be F and $30-F$

Then, $$F^2+(30-F)^2=650$$

Solving, we get $F=25$ N (or 5 N)

Therefore, the forces have magnitudes 25 N and 5 N

So, when the forces are applied at an angle 60° with each other, their resultant is

$$R=\sqrt{25^2+5^2+2(25)(5)\cos 60^\circ}=5\sqrt{31}\text{ N}$$

8.(B) Let the mass of each rod be m and let the length of each rod be L

$$I_1=\frac{1}{12}mL^2+2\left(\frac{1}{12}mL^2+m\left(\frac{L}{2}\right)^2\right)=\frac{3}{4}mL^2$$

$$I_2=3\left(\frac{1}{12}mL^2+m\left(\frac{L}{2\sqrt{3}}\right)^2\right)=\frac{1}{2}mL^2$$

So, $$I_1:I_2=\frac{3}{4}:\frac{1}{2}=3:2$$

9.(D) $$x(t)=A\sin^2\left(\omega t-\frac{\pi}{6}\right)=\frac{A}{2}\left(1-\cos\left(2\omega t-\frac{\pi}{3}\right)\right)$$

The general equation of SHM with mean position at $x=x_0$ is $x(t)=x_0+A\sin(\omega t+\phi)$

Therefore, by comparison, the mean position of the given SHM is at $x=\dfrac{A}{2}$

10.(A) At constant pressure, $\Delta Q=nC_P\Delta T \quad\Rightarrow\quad \Delta Q_1=n\left(\dfrac{5R}{2}\right)(120-20)$

At constant volume, $\Delta Q=nC_V\Delta T \quad\Rightarrow\quad \Delta Q_2=n\left(\dfrac{3R}{2}\right)(220-120)$

So, $$\frac{\Delta Q_2}{\Delta Q_1}=\frac{3}{5}\quad\Rightarrow\quad \Delta Q_2=\frac{3}{5}(\Delta Q_1)=60\text{ kJ}$$

11.(B) Fundamental frequency of an organ pipe open at both ends, $f_0=\dfrac{c}{2L}$

Fundamental frequency of an organ pipe closed at one end, $f=\dfrac{c}{4L'}=\dfrac{c}{4\left(\dfrac{2L}{3}\right)}=\dfrac{3}{4}f_0$

12.(D) Acceleration: $a(t)=Pt\,\hat{i}+Q\,\hat{j}$

Velocity: $v(t)=v_0+\int_0^t a(t)\,dt=\dfrac{1}{2}Pt^2\,\hat{i}+Qt\,\hat{j}$

Position: $r(t)=r_0+\int_0^t v(t)\,dt=\dfrac{1}{6}Pt^3\,\hat{i}+\dfrac{1}{2}Qt^2\,\hat{j}$

Now, if the particle passes through the point $(1,2)$, then

$$\frac{1}{6}Pt^3=1\qquad\text{and}\qquad\frac{1}{2}Qt^2=2$$

Both the above equations should give us the same value of t

Therefore, eliminating t, we get $P^2=\dfrac{9}{16}Q^3$

13.(B) At equator, $g' = g\left(1 - \frac{\omega^2 R}{g}\right)$

Also, $T = \frac{2\pi}{\omega} \Rightarrow$ As T decreases, ω increases

So the value of g' decreases

14.(B) The given body is a superposition of:

(i) a square lamina of mass m and side length a with its CM at $\left(\frac{a}{2}, \frac{a}{2}\right)$

(ii) a square lamina of mass $-\frac{b^2}{a^2}m$ and side length b with its CM at $\left(a - \frac{b}{2}, a - \frac{b}{2}\right)$

Therefore, the CM of the given body has coordinates:

$$X_{CM} = \frac{m\left(\frac{a}{2}\right) + \left(-\frac{b^2}{a^2}m\right)\left(a - \frac{b}{2}\right)}{m + \left(-\frac{b^2}{a^2}m\right)} = \frac{a^3 - b^2(2a - b)}{2(a^2 - b^2)} = \frac{a^3 + b^3 - 2ab^2}{2(a^2 - b^2)}$$

$$\Rightarrow \quad X_{CM} = \frac{a(a^2 - b^2) - b^2(a - b)}{2(a^2 - b^2)} = \frac{a(a + b) - b^2}{2(a + b)} = \frac{a^2 - b^2 + ab}{2(a + b)}$$

By symmetry, $Y_{CM} = X_{CM}$

So, the CM of the given body is at $\left(\frac{a^2 - b^2 + ab}{2(a + b)}, \frac{a^2 - b^2 + ab}{2(a + b)}\right)$

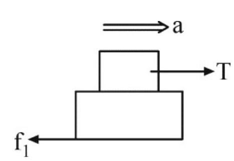
15.(A) Let the side length of the triangle be a

For equilibrium,

Clockwise torque due to F_1 = Anti-clockwise torque due to F_2

$$\Rightarrow \quad F_1\left(\frac{a}{\sqrt{3}}\right) = F_2\left(\frac{a}{2}\right) \quad \Rightarrow \quad 2F_1 = \sqrt{3}F_2$$

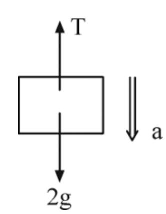
16.(B) From the combined force diagram of the two blocks,



$$T - 0.4g = 4a$$

And, from the force diagram of block C,

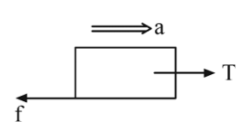
$$2g - T = 2a$$



Solving, we get $\quad a = \frac{4g}{15} \quad$ and $\quad T = \frac{22g}{15}$

Now, from the force diagram of block A,

$$T - f = 3a \quad \Rightarrow \quad f = T - 3a = \frac{2g}{3}$$



But $f \le \mu(3g) \Rightarrow \quad \mu \ge \frac{2}{9}$

17.(C) Let the radius of curvature be R

We know that the height of capillary rise is given by the pressure equation $h\rho g = \dfrac{2\sigma}{R}$

When the capillary length is sufficient, the meniscus makes the natural angle of contact with the capillary and the natural height of capillary rise, h, is obtained from the above equation with $R = \dfrac{r}{\cos\theta}$. But, when the length is insufficient (let it be h_2), the liquid rises to the top, and then the shape of the meniscus changes, i.e. its radius of curvature changes

$$h_2\rho g = \frac{2\sigma}{R'}$$

As $h_2 < h \Rightarrow R' > R$

18.(B) Velocity: $v = at$

Power: $P = Fv = (ma)(at)$

Therefore, the instantaneous power is proportional to t

Distance: $s = \dfrac{1}{2}at^2$

Since the distance s is proportional to t^2, the instantaneous power is proportional to $s^{1/2}$

19.(D) After the isothermal expansion, the pressure is $\dfrac{P_0}{2}$, and the temperature is T_0

Now, for the adiabatic process,

$$PV^{5/3} = \text{constant} \quad \Rightarrow \quad P_f = \left(\frac{2V_0}{V_0}\right)^{5/3}\left(\frac{P_0}{2}\right) = 2^{2/3}P_0$$

And, the final temperature, $T_f = \left(\dfrac{P_f}{P_0}\right)T_0 = 2^{2/3}T_0$

20.(A) Assuming that there is no slipping between the blocks, they oscillate with angular frequency $\omega = \sqrt{\dfrac{k}{2m}}$

Friction acting on the upper block plays the role of restoring force ($m\omega^2 x$, where x is the distance from the mean position) necessary to keep the upper block in SHM. We know that the maximum magnitude of friction that can act on the upper block is $f_{\max} = \mu mg$. Therefore, there is no slipping between the blocks if the maximum magnitude of the restoring force during SHM is less than $f_{\max}$.

So, $$m\omega^2 L \le \mu mg \quad \Rightarrow \quad L \le \frac{2\mu mg}{k}$$

SECTION-2

21.(70) Let the final temperature be T°C

Then, Heat gained by ice and melted water = Heat lost by water

$$\Rightarrow \quad 10(336) + 10(4.2)(T-0) = 150(4.2)(80-T)$$

Solving, we get $T = 70$

22.(8) Young's Modulus $= \dfrac{\text{Stress}}{\text{Strain}}$ and Stress $= \dfrac{F}{\left(\dfrac{\pi d^2}{4}\right)}$

$$\frac{Y_P}{Y_Q} = \left(\frac{Strain_Q}{Strain_P}\right)\left(\frac{2d}{d}\right)^2 = \left(\frac{0.02}{0.01}\right)(4) = 8$$

23.(36) Required time $= \dfrac{\text{Distance between B and the man}}{\text{Velocity of B relative to the man}}$ $\Rightarrow$ $t = \dfrac{100}{(15-5)\left(\dfrac{5}{18}\right)} = 36 \text{ s}$

24.(15) We know that the efficiency, $\eta = 1 - \dfrac{Q_{out}}{Q_{in}}$

Therefore, $Q_{out} = Q_{in}(1-\eta) = 20(1-0.25) = 15 \text{ kJ}$

25.(5) Observed frequency, $f = f_0\left(\dfrac{c - v_O}{c - v_S}\right) = f_0\left(\dfrac{c-(-0.03c)}{c - 0.02c}\right) = f_0(1+0.03)(1-0.02)^{-1} = (1+0.05)f_0$

Chemistry

SECTION-1

1.(D) XeF_5^+ = Square pyramidal

SiF_5^- = Trigonal bipyramidal

AsF_4^+ = Tetrahedral

ICl_4^- = Square planar

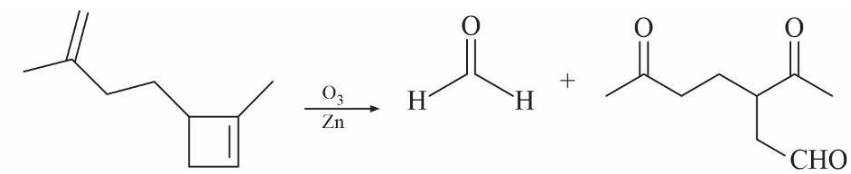
2.(D) $Na \longrightarrow 1s^2 2s^2 2p^6 3s^1$

$3s^1$ orbitals has $n = 3, \ell = 0$

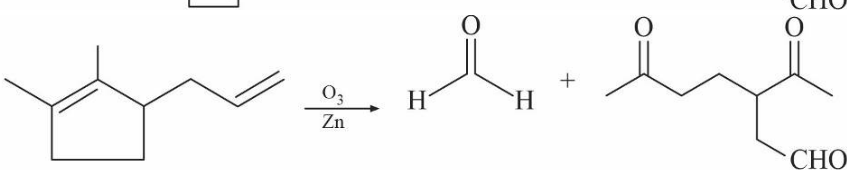
Number of radial nodes $= n - \ell - 1 = 3 - 0 - 1 = 2$

3.(B) $\frac{r_{O_2}}{r_{CH_4}} = \frac{P_{O_2}}{P_{CH_4}}\sqrt{\frac{M_{CH_4}}{M_{O_2}}} = \frac{n_{O_2}}{n_{CH_4}}\sqrt{\frac{16}{32}} = \frac{3}{32} \times \frac{16}{2}\sqrt{\frac{16}{32}} = \frac{3}{4\sqrt{2}}$

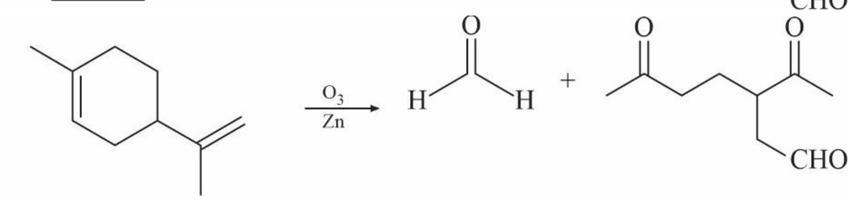
4.(D) (A)



(B)



(C)



5.(B) Optimum value of BOD is 5 ppm.

6.(B)

$$Al(s) \quad + \quad \frac{3}{2}Cl_2(g) \quad \rightleftharpoons \quad AlCl_3(s)$$

At equilibrium $\quad - \quad p' \quad 13.35\,gm \equiv 0.1\,mole$

p' = pressure of the remaining chlorine gas at equilibrium.

Initial moles of $Cl_2(g) = \frac{PV}{RT} = \frac{5 \times 5}{25} = \frac{25}{25} = 1\,mole$

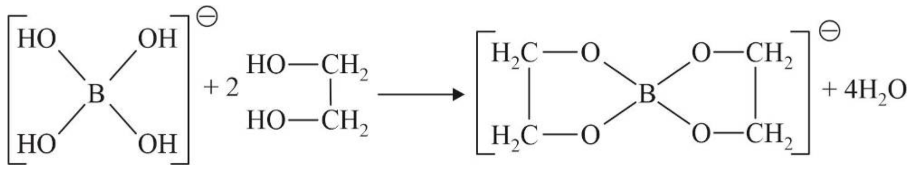
Final moles of $Cl_2(g) = 1 - \frac{3}{2}[0.1] = 1 - 0.15 = 0.85\,mole$

Final pressure of $Cl_2(g) = \frac{0.85[50]}{5} = 8.5\,atm = p'$

$$K_p = \frac{1}{(p')^{3/2}} = (p')^{-3/2} = (8.5)^{-3/2}$$

7.(C) **(A)** By Le-chatelier principle, adding additional reactant to a system will shift the reaction to right.

(B) On adding ethylene glycol, they form complexing species with orthoboric acid.

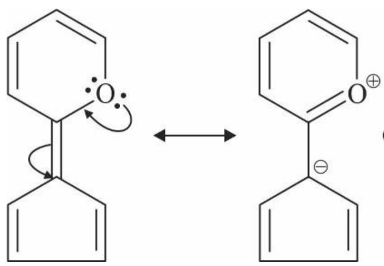


8.(C) Above 500 ppm of SO_4^{2-} ions in drinking water can cause laxative effect.

Maximum limit of NO_3^- ions in drinking water is 50 ppm, above this limit it can cause the decrease like methemoglobinemia.

More than 1 ppm F^- ions in drinking water are not fit for drinking, it can cause decay of bones and teeth.

9.(B)



10.(B) According to Fazan's rule,

Covalent character $\propto Z^+$

$\propto Z^-$

$\propto \frac{1}{r^+}$

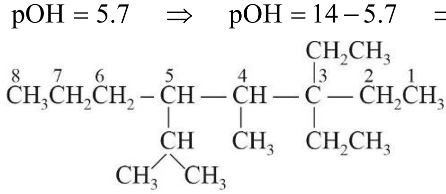
$\propto r^-$

11.(A) $pOH = pK_b + \log\frac{[NH_4Cl]}{[NH_4OH]}$

$pOH = 4.7 + \log\left(\frac{0.1}{0.01}\right)$

$pOH = 5.7 \quad \Rightarrow \quad pOH = 14 - 5.7 \quad \Rightarrow \quad 8.3$

12.(A)



3, 3 - Diethyl-4-methyl-5-isopropyloctane

13.(B) $C \rightarrow B$ part in the graph represents fusion at a constant temperature of 600 K and for 40 min.

Time of fusion $= 45 - 5 = 40\,\text{min}$

So, total heat exchange during the fusion $= \frac{0.4 \times 10^3 \times 40}{2}\text{ J mol}^{-1}$

$\therefore \qquad \Delta S_{\text{fusion}} = \frac{0.4 \times 10^3 \times 40}{2 \times 600} = 13.33\text{ J mol}^{-1}\text{K}^{-1}$

Hence (B) is correct.

14.(C) $\Delta H = \Delta U + \Delta(PV) = \Delta U + P_2V_2 - P_1V_1$

We cannot use, $\Delta H = \Delta U + \Delta nRT$ as $PV \neq nRT$ since gases deviate from ideal behaviour.

$\therefore \quad P_2V_2 - P_1V_1 = 40 \times 1 - 70 \times 1 = -30$ lit atm $= -3$ kJ

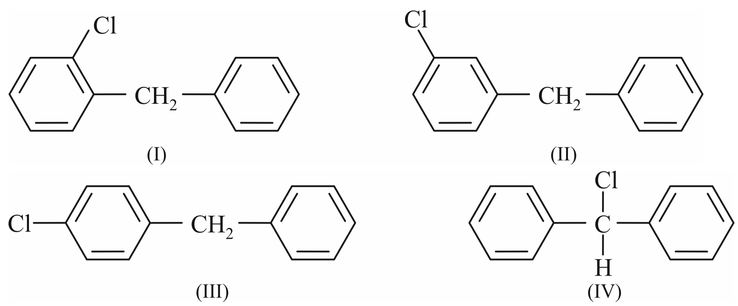
Also, $\quad -560 = \Delta U - 3.0$

$\Delta U = -557$ kJ

15.(B) Higher the concentration of common ion, the smaller is the solubility.

16.(C) It is disproportionation reaction wrt. phosphorous.

17.(D)



18.(A)

SO_2Cl_2	+	$4NaOH$	$\rightarrow$ Na_2SO_4	+	$2NaCl + 2H_2O$
$\text{moles} = \dfrac{27000}{135} = 200$			200 moles		400 moles
			$m = 200 \times 142g$		$m = 400 \times 58.5$ g
			$= 28.400$ kg		$= 23.4$ kg

$\Rightarrow x = 28.4 - 23.4 = 5kg$

19.(C) On moving left to right ionization energy increases as size decreases and on moving top to bottom in a group, size increases. So ionization energy decreases. Hence order of increasing first ionization enthalpy is $Ba < Ca < Se < S < Ar$.

20.(A) Chlorine is simultaneously reduced and oxidised.

SECTION-2

21.(5) Alkali metal and alkaline earth metals can be identified by flame test.

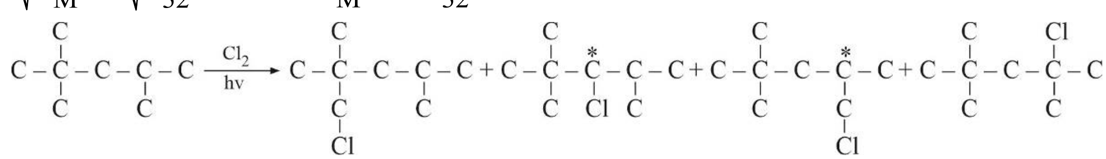
22.(3) $v = 2.18 \times 10^6 \dfrac{Z}{n}$ m/s

For $Li^{2+}, Z = 3$; $\quad$ So, $n = 3$

as $\quad v = 2.18 \times 10^6$ m / s

23.(8) $\sqrt{\dfrac{2RT_1}{M}} = \sqrt{\dfrac{3RT_2}{32}} \quad \Rightarrow \quad \dfrac{2 \times 150}{M} = \dfrac{3 \times 400}{32} \quad \Rightarrow \quad M = 8$

24.(6)



25.(4) $k = \dfrac{R}{N_A} \quad \therefore \quad R = k \cdot N_A = 6.023 \times 10^{23} \times 1.380 \times 10^{-23} J \cdot mol^{-1} \cdot k^{-1}$

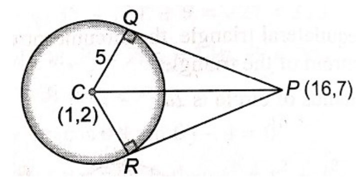
There are 4 significant figures in each term. Hence, there will be 4 significant digits in the calculated value of R.

Mathematics

SECTION-1

1.(D) Given equation of circle is $x^2 + y^2 - 2x - 4y - 20 = 0$

$\therefore$ Centre is (1, 2) and radius, $r =^2 + 2^2 + 20 = 5$



Now, $PC = \sqrt{(16-1)^2 + (7-2)^2} = \sqrt{250}$

In ΔPCQ,

$$PQ = \sqrt{PC^2 + QC^2} = \sqrt{\left(\sqrt{250}\right)^2 - (5)^2} = 15$$

$\therefore$ Area of quadrilateral PQCR

$$= 2 \cdot \text{area of } \Delta PCQ = 2 \cdot \frac{1}{2} PQ \cdot QC = 1 \cdot 15 \cdot 5 = 75 \text{ sq unit}$$

2.(A) The slope of line $x - 2y = 3$ is $\frac{1}{2}$

Let the slope of required lines is m.

$$\therefore \quad \tan 45^\circ = \pm \left| \frac{\frac{1}{2} - m}{1 + \frac{m}{2}} \right| \Rightarrow 1 + \frac{m}{2} = \pm\left(\frac{1}{2} - m\right) \Rightarrow m = -\frac{1}{3}, 3$$

$\therefore$ Equation of line with slope $m = -\frac{1}{3}$ and passing through (3, 2), is

$$(y-2) = -\frac{1}{3}(x-3) \Rightarrow x + 3y = 9$$

And another equation of line with slope $m = 3$ and passing through (3, 2), is

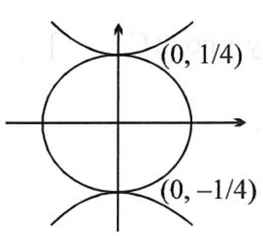
$$(y-2) = 3(x-3) \Rightarrow x + 3y = 7$$

3.(A) $\because$ 10^{th} term in the expansion of $(2-3x^3)^{20}$ is ${}^{20}C_9 2^9 (-1)^9 (2)^{11} (3)^9 x^{27}$

and 11^{th} term is ${}^{20}C_{10} 2^{10} 3^{10} x^{30}$

$$\therefore \quad \frac{{}^{20}C_9 (-1)^9 (2)^{11} (3)^9 x^{27}}{{}^{20}C_{10} \cdot 2^{10} \cdot 3^{10} \cdot x^{30}} = \frac{45}{22} \Rightarrow -\frac{10}{11} \cdot \frac{2}{3x^3} = \frac{45}{22} \Rightarrow x^3 = -\frac{8}{27} \Rightarrow x = -\frac{2}{3}$$

4.(D) $\dfrac{y^2}{\frac{1}{16}} - \dfrac{x^2}{\frac{1}{9}} = 1$



Locus will be the auxiliary circle $x^2 + y^2 = \frac{1}{16}$

5.(C) A number is divisible by 3, if the sum of the digits is divisible by 3.

Since, $1+2+3+4+5=15$ is divisible by 3, therefore total such numbers is $5!$ *i.e.*, 120.

And,. Other five digits whose sum is divisible by 3 are 0, 1, 2, 4, 5.

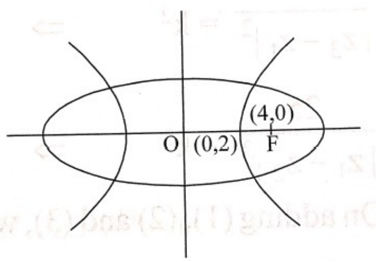
Therefore, number of such formed numbers $=5!-4!=96$

Hence, the required number of numbers $=120+96=216$

6.(B) For the ellipse, $a^2=25, b^2=9$

$\therefore \quad 9=25(1-e^2) \Rightarrow e^2=\frac{16}{25} \Rightarrow e=\frac{4}{5}$

$\therefore$ One of the foci is $(ae,\ 0)$ i.e. (4, 0)



$\therefore \quad a'e'=4 \Rightarrow \sqrt{2}a'=4 \Rightarrow a'=2\sqrt{2}$ and $b'^2=4(e'^2-1)=8$

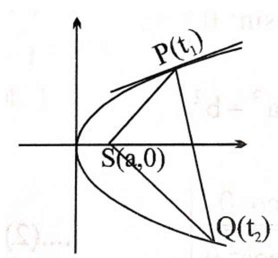
$\therefore$ equation of the hyperbola is $\frac{x^2}{8}-\frac{y^2}{8}=1$

7.(D) $at_1^2=2at_1 \Rightarrow t_1=2$; $P(16, 16)$

$t_2=-t_1-\frac{2}{t_1}=-3$

$\therefore \quad Q(36, 24)$

$\therefore \quad \left.\begin{array}{l} m_{SP}=\frac{16}{16-4}=\frac{4}{3} \\ m_{SQ}=\frac{-24}{36-4}=-\frac{3}{4} \end{array}\right] \angle PSQ=90^\circ$



8.(D) $AB=\sqrt{a^2+b^2}$

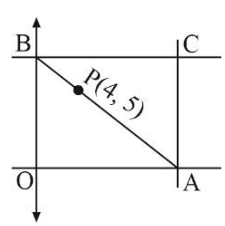
Hence, $D=\sqrt{b^2+a^2}$

9.(C) Equation of any line through $P(4,\ 5)$ with slope m will be

$y-5=m(x-4) \Rightarrow mx-y=4m-5$

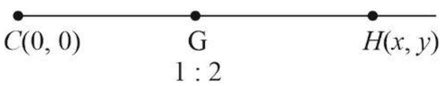
Clearly $A\left(\frac{4m-5}{m},\ 0\right)$ and $(0,\ 5-4m) \equiv C(h,\ k)$

Now $\frac{k}{h}=-m \quad \Rightarrow \quad m=\frac{-k}{h} \Rightarrow k=5-4m=5-4\left(\frac{-k}{h}\right)$



$\therefore$ Locus of $C(h,\ k)$ is $y=5+\frac{4y}{x} \Rightarrow \frac{4}{x}+\frac{5}{y}=1$

10.(C)



$$\left(\frac{\sqrt{3}-2\cos\theta+2\sin\theta}{3}, \frac{1+2\sin\theta+2\cos\theta}{3}\right)$$

$$\frac{x}{3}=\frac{\sqrt{3}-2\cos\theta+2\sin\theta}{3} \Rightarrow x=1-2\cos\theta+2\sin\theta$$

$$\frac{y}{3}=\frac{1+2\sin\theta+2\cos\theta}{3} \Rightarrow y=1+2\sin\theta+2\cos\theta$$

$$(x-\sqrt{3})^2+\left(y-1\right)^2=8$$

11.(D) $\sin^3 x+\sin^4 x+\sin^5 x+.....$

$$f(x)=\frac{\sin^3 x}{1-\sin x}$$

$$f\left(\frac{\pi}{3}\right)=\frac{3\sqrt{3}}{8-4\sqrt{3}}$$

12.(B) We have, $x^2+x-1<^{\alpha}_{\beta}$

$\therefore \quad \alpha+\beta=-1\,;\ \alpha\beta=-1$

$$\frac{\alpha\beta^4(\beta+1)^4+\beta\alpha^4(\alpha+1)^4}{\alpha^2+\beta^2+\alpha+\beta}=\frac{\alpha+\beta}{(\alpha+\beta)^2-2\alpha\beta+\alpha+\beta}=-\frac{1}{2}$$

13.(B) $\sin x=\sin 2y$ …………(i), $\cos x=\sin y$ ………..(ii)

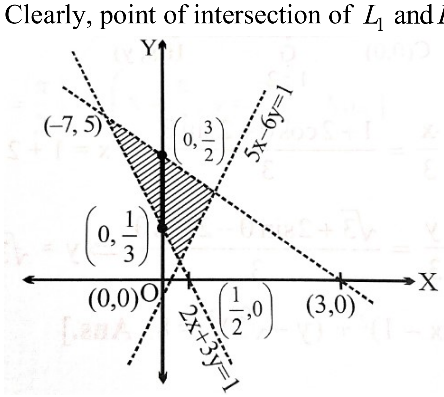
$\therefore \quad (\text{i})^2+(\text{ii})^2 \Rightarrow 1=\sin^2 y(1+4\cos^2 y)$

$\therefore \quad \cos^2 y=4\sin^2 y\cdot\cos^2 y \Rightarrow \cos^2 y(4\sin^2 y-1)=0$

$\therefore \quad y=\frac{\pi}{2}$ or $\frac{\pi}{6}$

$$\left.\begin{array}{l}\text{if } y=\frac{\pi}{2} \text{ then } x=0\\ \text{if } y=\frac{\pi}{6} \text{ then } x=\frac{\pi}{3}\end{array}\right] \Rightarrow 2 \text{ ordered pairs. i.e., } \left(x=0,\ y=\frac{\pi}{2}\right) \text{ or } \left(x=\frac{\pi}{3},\ y=\frac{\pi}{6}\right)$$

14.(B) Clearly, point of intersection of L_1 and L_2, is $(0, -k)$ which lies on y-axis.



Clearly, from figure, we get $k\in\left(-\frac{3}{2}, -\frac{1}{3}\right)$

15.(D) $S = 29\binom{30}{0} + 28\binom{30}{1} + 27\binom{30}{2} + ... + 1\binom{30}{28} + 0\binom{30}{29} - \binom{30}{30}$

Also $\quad S = -\binom{30}{0} + 0\binom{30}{1} + 1\binom{30}{2} + ... + 27\binom{30}{28} + 28\binom{30}{29} + 29\binom{30}{30}$

$\Rightarrow \quad 2S = 28\left({}^{30}C_0 + {}^{30}C_1 + {}^{30}C_2 + ... + {}^{30}C_{30}\right) = 28(2^{30})$

Hence, $S = 14 \times 2^{30} = \frac{7}{2} \times 2^{32} \Rightarrow k = \frac{7}{2}$

16.(C) 1st place is seven + when 2nd place is seven +7 is not in number

$\boxed{7}\boxed{8}\overset{5}{\boxed{\;}} + \overset{8}{\boxed{\;}}\boxed{7}\boxed{8} + \underset{8}{\boxed{\;}}\underset{9}{\boxed{\;}}\underset{5}{\boxed{\;}}$

$5 + 8 + 360 = 373$

17.(D) Given, $x^2 - \sqrt{3}x + 1 = 0$

$\Rightarrow \quad x = \frac{\sqrt{3} \pm \sqrt{3-4}}{2} = \frac{\sqrt{3} \pm i}{2} = \cos\frac{\pi}{6} \pm i\sin\frac{\pi}{6}$

$\Rightarrow \quad x^n = \cos\frac{n\pi}{6} \pm i\sin\frac{n\pi}{6}$ and $\frac{1}{x^n} = \cos\frac{n\pi}{6} \mp i\sin\frac{n\pi}{6}$

$\therefore \quad x^n - \frac{1}{x^n} = \pm 2i\sin\frac{n\pi}{6} \quad \Rightarrow \quad \left(x^n - \frac{1}{x^n}\right)^2 = -4\sin^2\frac{n\pi}{6}$

Hence, $\sum_{n=1}^{24}\left(x^n - \frac{1}{x^n}\right)^2 = -\left[\sin^2\frac{\pi}{6} + \sin^2\frac{2\pi}{6} + ... + \sin^2\frac{24\pi}{6}\right] = -4(12) = -48$

18.(D) As, $\quad S_{25} = \left\{\frac{26}{25}, \frac{51}{25}, \frac{76}{25}, ...\text{upto 25 terms}\right\}$

Here, $\quad a = \frac{26}{25},\ n = 25,\ d = 1 \qquad \therefore \qquad S_{25} = \frac{25}{2}\left(\frac{52}{25} + 24\right) = 326$

19.(B) Number of friends to be invited = 6

Let A, B be the friends who are not to attend the party together. Either none of A, B or one of A, B attend the party

$\therefore$ Number of ways of inviting friends

$= {}^{10-2}C_6 \times {}^2C_0 + {}^{10-2}C_5 \times {}^2C_1$

$= 25 \times 1 + 56 \times 2 = 140$

20.(B) Any element, is of the form of $\frac{6!}{t_1!\,t_2!\,t_3!\,t_4}\left(x^2\right)^{t_1}\left(\frac{1}{x^2}\right)^{t_2}(y)^{t_3}\left(\frac{1}{y}\right)^{t_4}$

Where $t_1 + t_2 + t_3 + t_4 = 6$; $t_1 \geq 0$

The constant term occur when $t_1 = t_2$ and $t_3 = t_4$

So, $t_1 + t_3 = 3 \Rightarrow (0, 3)$; $(3, 0)$; $(1, 2)$, $(2, 1)$ $\Rightarrow$ constant term:

$2\left(\frac{6!}{0!\,0!\,3!\,3!} + \frac{6!}{2!\,2!\,1!\,1!}\right) = 400$

SECTION-2

21.(14) We have, $1-\log_3|x+1|=\log_3\left(\frac{x+5}{x+3}\right)$

$\Rightarrow \quad \frac{3}{|x+1|}=\frac{x+5}{x+3}$ …………..(1)

Now, we shall consider two cases:

Case I: When $x+1>0 \Rightarrow x>-1$

Now, $3(x+3)=(x+1)(x+5)$

$\Rightarrow \quad x^2+3x-4=0 \begin{cases} x=-4 \\ \text{or} \\ x=1 \end{cases}$

But $x=-4$ (rejected)

$\Rightarrow \quad x=1$

Case II:

When $x+1<0 \Rightarrow x<-1$

Now, $3(x+3)=-(x+1)(x+5) \Rightarrow \quad x^2+9x+14=0 \begin{cases} x=-2 \\ \text{or} \\ x=-7 \end{cases}$

Hence, solution set = $\{-7, -2, 1\}$

So, number of values of x satisfying the given equation are 3

22.(6) $g_2g_{n-1}=128=g_1g_n$ (Using property of G.P.)

$\Rightarrow \quad g_1+g_n=66$ ………..(i)

$\therefore \quad g_n-g_1=\sqrt{(g_1+g_n)^2-4g_1g_n}=\sqrt{(66)^2-4\times128} \Rightarrow g_n-g_1=62$ …….(ii)

$\therefore$ From (i) & (ii), we get $g_n=64,\ g_1=2$

Now, let common ratio of G.P. be r, so

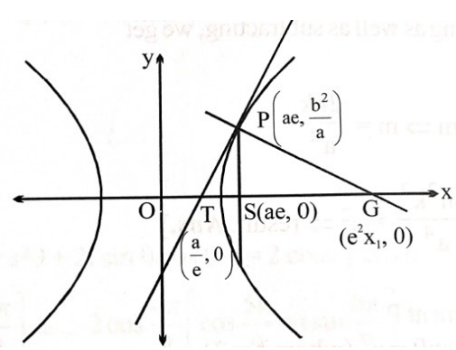
$g_n=g_1r^{n-1}=2\cdot r^{n-1}=64 \quad \Rightarrow \quad r^{n-1}=32$ ……….(iii)

Now, $\sum_{i=1}^{n} g_i=126 \Rightarrow \frac{2(r^n-1)}{(r-1)}=126 \Rightarrow \frac{r^n-1}{r-1}=63 \Rightarrow 32r-1=63(r-1)$ [on using (iii)]

$\Rightarrow \quad 31r=62 \quad \therefore \quad r=2$ ………..(iv)

Hence from (iii) & (iv), we get $2^{n-1}=32=2^5 \Rightarrow n=6$

23.(45) $P\left(ae, \frac{b^2}{a}\right)$



$T: \frac{xx_1}{a^2}-\frac{yy_1}{b^2}=1$

$\frac{x\ a\ e}{a^2}-\frac{y\ b^2}{b^2a}=1\ ;\quad \frac{ex}{a}-\frac{y}{a}=1$

Put $y=0$ to get

$T=\left(\frac{a}{e}, 0\right)$

$N : \dfrac{a^2 x}{ae} + \dfrac{b^2 y}{y_1} = a^2 + b^2$; put $y = 0$ to get

$G = \dfrac{e(a^2 + b^2)}{a} = (ae^3, 0)$

$G : (ae^3, 0)$

$\text{Area} = \left(ae^3 - \dfrac{a}{e} \right) \cdot \dfrac{b^2}{a} \cdot \dfrac{1}{2} = \dfrac{(e^4 - 1)b^2}{2e}$ Now $e^2 = 1 + \dfrac{12}{4} = 4 \Rightarrow e = 2$ and $b^2 = 12$

$\text{Area} = \dfrac{15 \cdot 12}{2 \cdot 2} = 45$

24.(65) $x^4 + ax^3 + bx^2 + cx + d = 0 \left\langle \begin{array}{l} \alpha \\ \beta \\ \gamma \\ \delta \end{array} \right.$

Given $\alpha + \beta = 4 + 5i$ …………(1) (say)

And $\gamma\delta = 12 + i$ …………(2) (say)

Since four roots must occur in pair of conjugate complex, hence $(\alpha \text{ and } \beta)$ and $(\gamma \text{ and } \delta)$ can not be complex conjugate since their sum and product is not real.

$\therefore$ Suppose $\alpha = \overline{\gamma}$

and $\beta = \overline{\delta}$

$\alpha + \beta = \overline{\gamma} + \overline{\delta}$

$\therefore$ $\overline{\alpha} + \overline{\beta} = \gamma + \delta$

$4 - 5i = \gamma + \delta$ …………..(3) (from (1))

Similarly

$\alpha\beta = \overline{\gamma}\ \overline{\delta}$

$\alpha\beta = 12 - i$ (from 2)

Now $b = \Sigma\alpha\beta = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = (\alpha\beta + \gamma\delta) + (\alpha + \beta)(\gamma + \delta)$

$= (12 - i) + (12 + i) + (4 + 5i)(4 - 5i) = 24 + 41 = 65$

25.(28) $R + B + G = 6 \Rightarrow {}^8C_2 = 28$